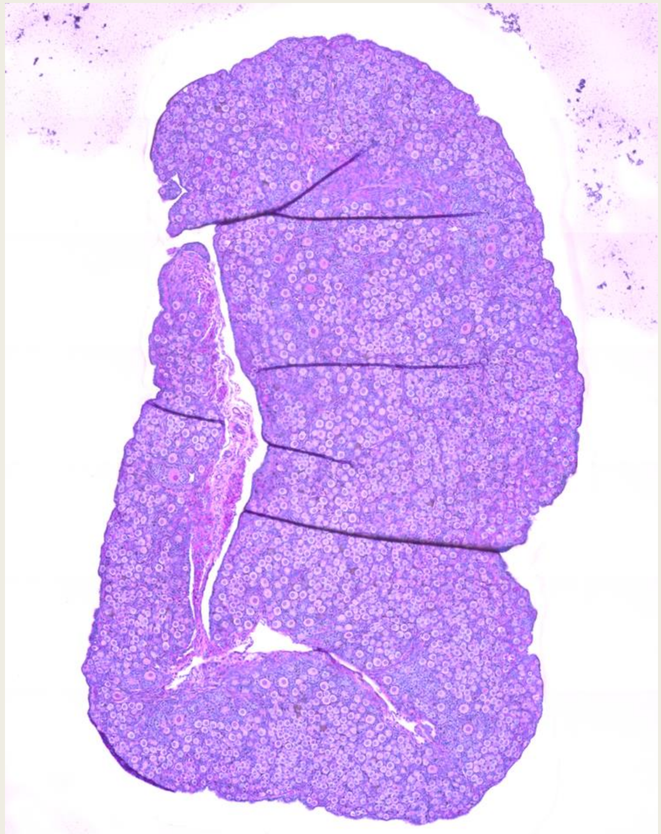
Machine Learning-based Quantification Of Ovarian Follicle Stages In *Heterocephalus Glaber*

Julian Coles, Martin Orkuma, Pamela Styborski, and Silvia Tenempaguay-Nunez

New College of Interdisciplinary Arts and Sciences,
Arizona State University, Tempe, Arizona

Introduction



Manual ovarian follicle identification and quantification is labor-intensive, observer-dependent, and limits the scale of reproductive histology research.

Why now:

1. Histology is going digital: Convolutional neural networks work well now
2. Reproducibility is the main bottleneck
3. MOTHER fills the gap and provides the right kind of data
4. *H. glaber* is the right organism - and no one's built a tool for it yet

Ready-to-use slide repositories, pretrained CNN models, and an understudied species
now make it possible to build a focused, reproducible tool for follicle classification.

Research Questions

Can a machine-learning pipeline replace manual follicle counting in *Heterocephalus glaber* ovarian histology?

The gap:

- Manual follicle identification and quantification is labor-intensive and limits the scale of ovarian histology research.
- The naked mole rat is a unique model for reproductive longevity, yet automated tools for its ovarian architecture are lacking.
- The MOTHER database provides FAIR-aligned WSIs, but no reproducible ML workflow exists for *H. glaber*.

Can a ResNet34 model, trained on annotated tiles from QuPath, accurately classify follicle stages in naked mole-rats and aggregate those predictions into reliable slide-level counts?

Hypothesis

Machine learning can automate the tasks of identifying, classifying, and quantifying ovarian follicles in *H. glaber*, replacing manual annotation with a reproducible, scalable pipeline.

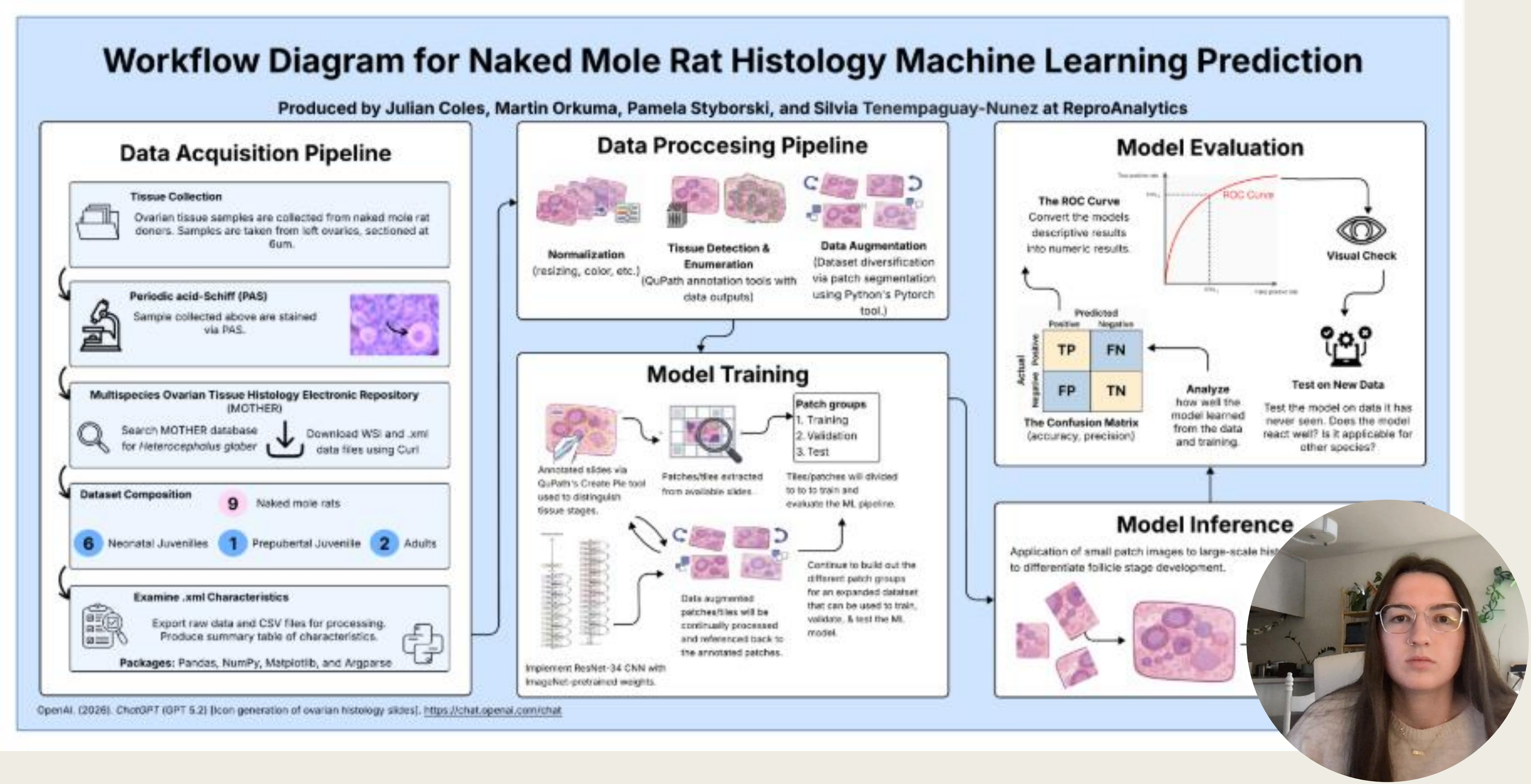
What this means in practice:

- *Identifying* → detect tissue regions in each whole-slide image
- Classifying → assign each tile to one of eight follicle classes
- *Quantifying* → aggregate tile-level predictions into slide-level follicle counts
- *Automating* → run the full pipeline end-to-end without manual intervention after annotation

[WSI] → [tile grid] → [labeled tiles] → [count bars]

detect classify aggregate report

Automated Workflow Pipeline

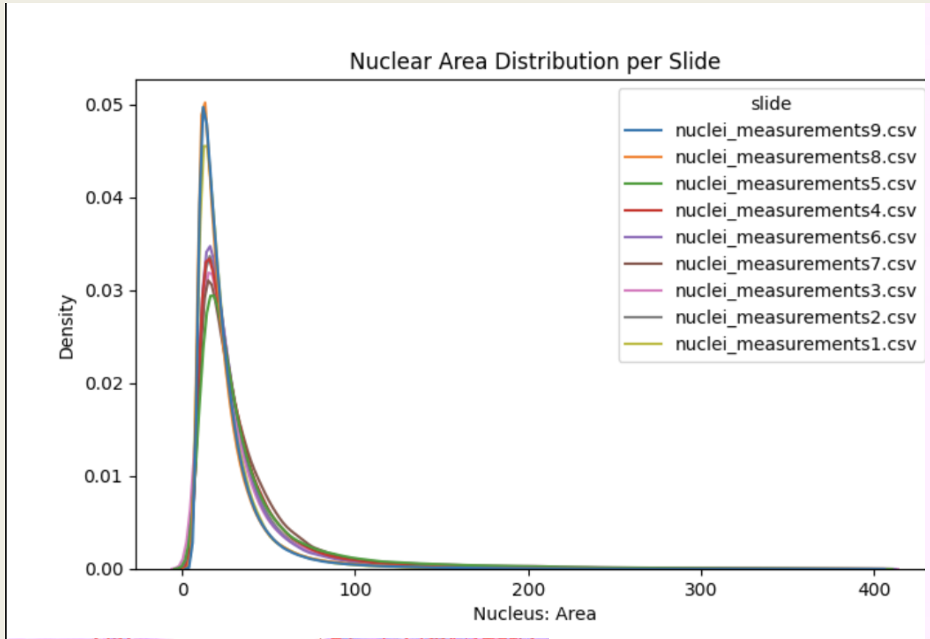


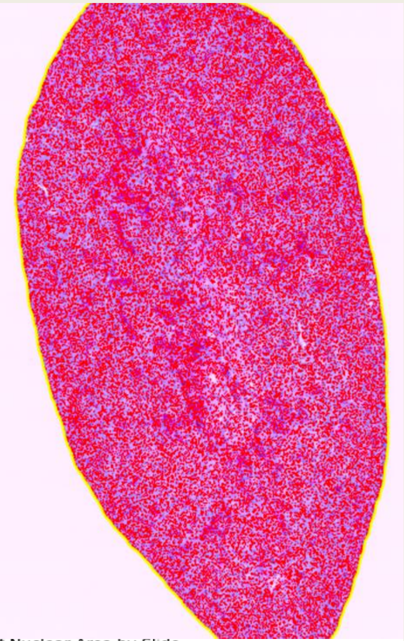
Data Source and Acquisition

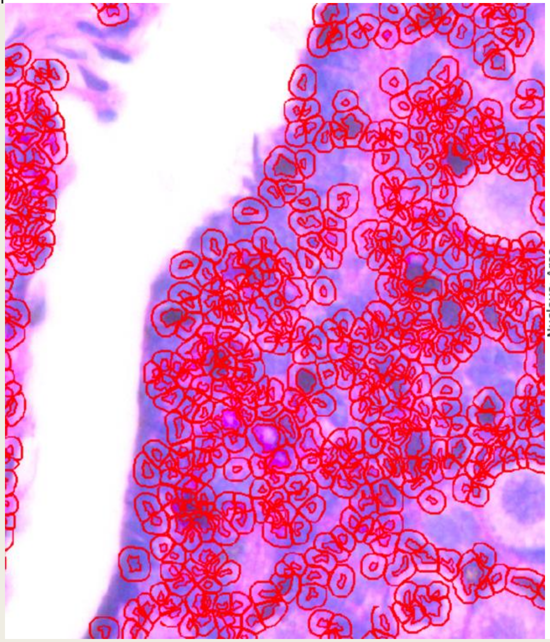
	Source	Common Name	Scientific Name	Donor ID	Donor Life Stage	Donor Age (Years)	Donor Age (Days)	Donor Sex	Ovary Sampled
0	MDB0000530-P28-Ova-106_s55.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P28-Ova-106	neonatal	0	28	female	left
1	MDB0000531-P1-11-s40.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P1-Ova11	neonatal	0	1	female	left
2	MDB0000532-P15_F09OV-s83.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P15-F09	neonatal	0	15	female	left
3	MDB0000533-P180_6Mon-6b-s86.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_NMR-6Mon-6B	prepubertal	0	180	female	left
4	MDB0000534-P5-23-s13.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P5-Ova23	neonatal	0	5	female	left
5	MDB0000535-P8-F02-OV-s135.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P8-F02	neonatal	0	8	female	left
6	MDB0000536-P90-NMR_3Mo-3B-s96.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P90-3B	neonatal	0	90	female	[illegible]
7	MDB0000537-2653-3y-Exsub-s19.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_2653	adult	3	0	fem	[illegible]
8	MDB0000538-2659-3y-Sub-s109.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_2659	adult	3	0	[illegible]	[illegible]

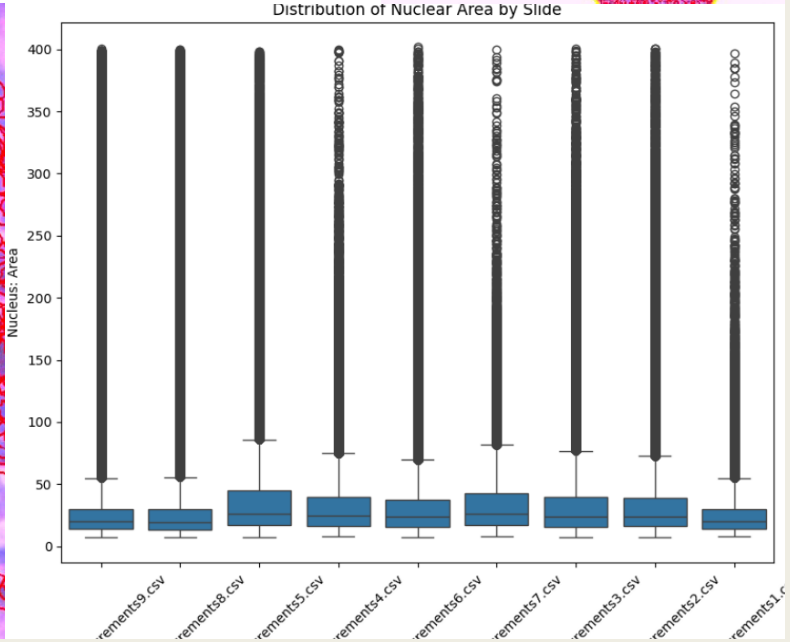
Exploratory Data Analysis

- EDA QuPath Annotations
- QuPath measurements and Kernel density estimation (KDE) plots characterized slide variability
- Boxplot emphasized similar trend as KDE



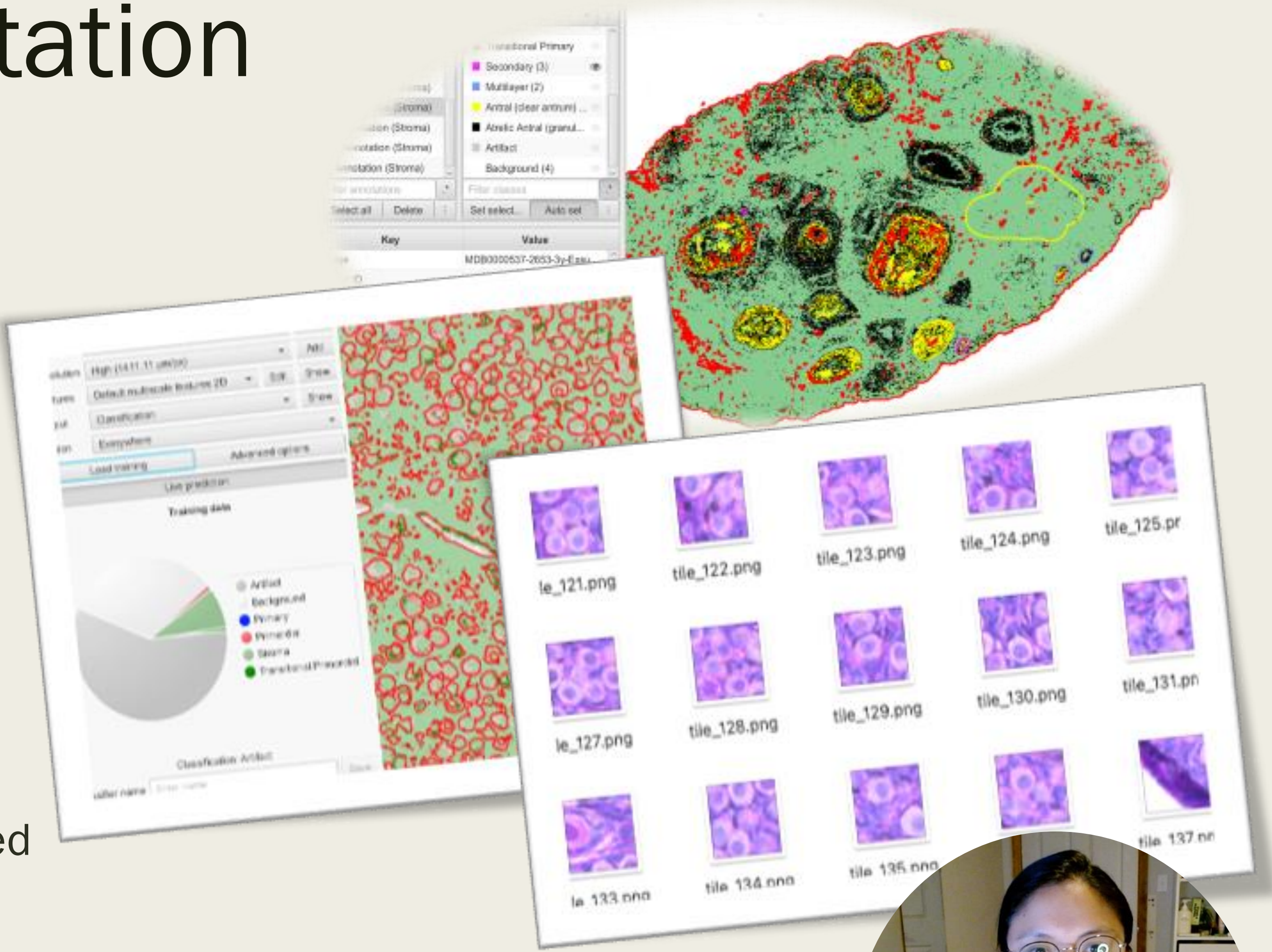


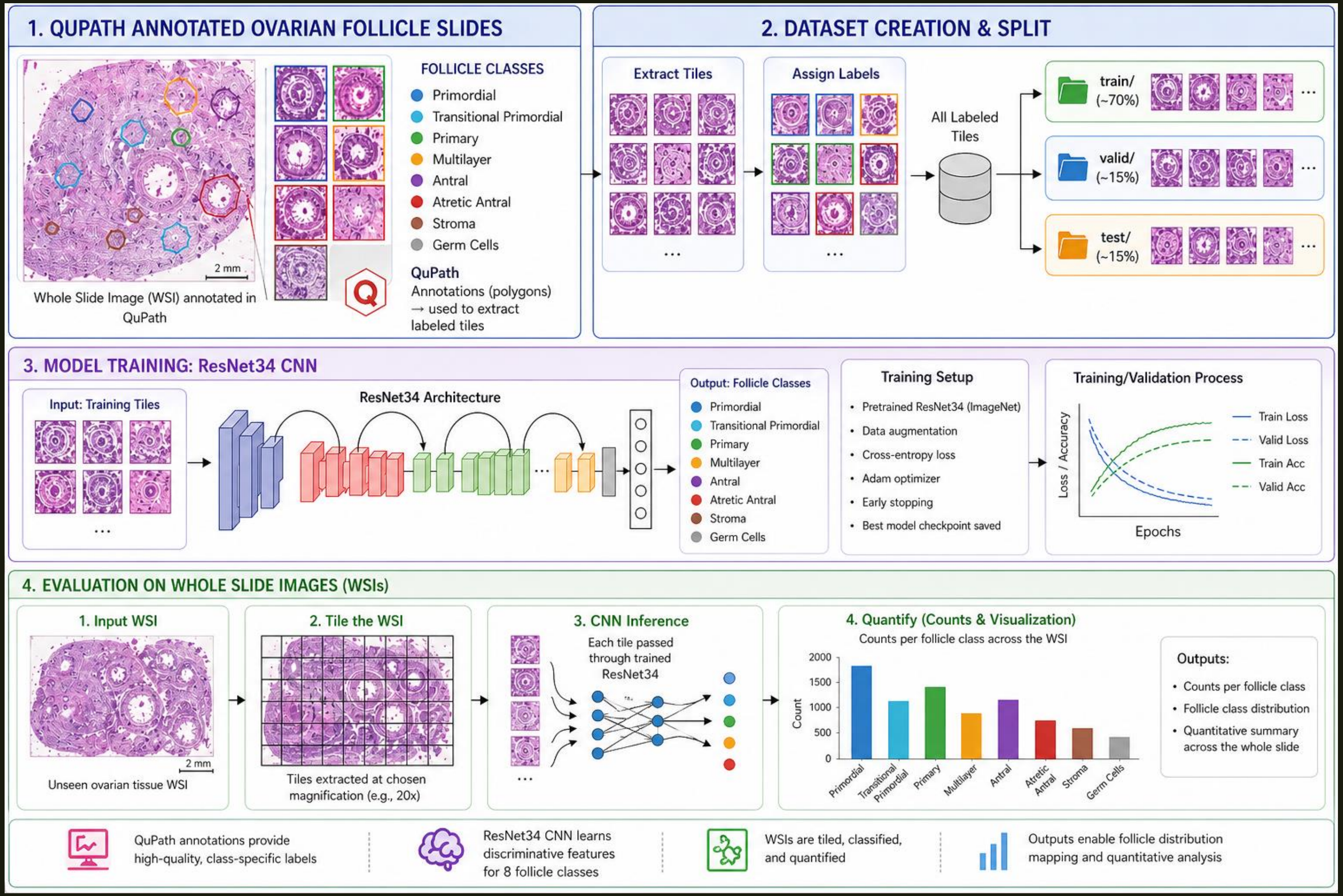




Methodology: Annotation

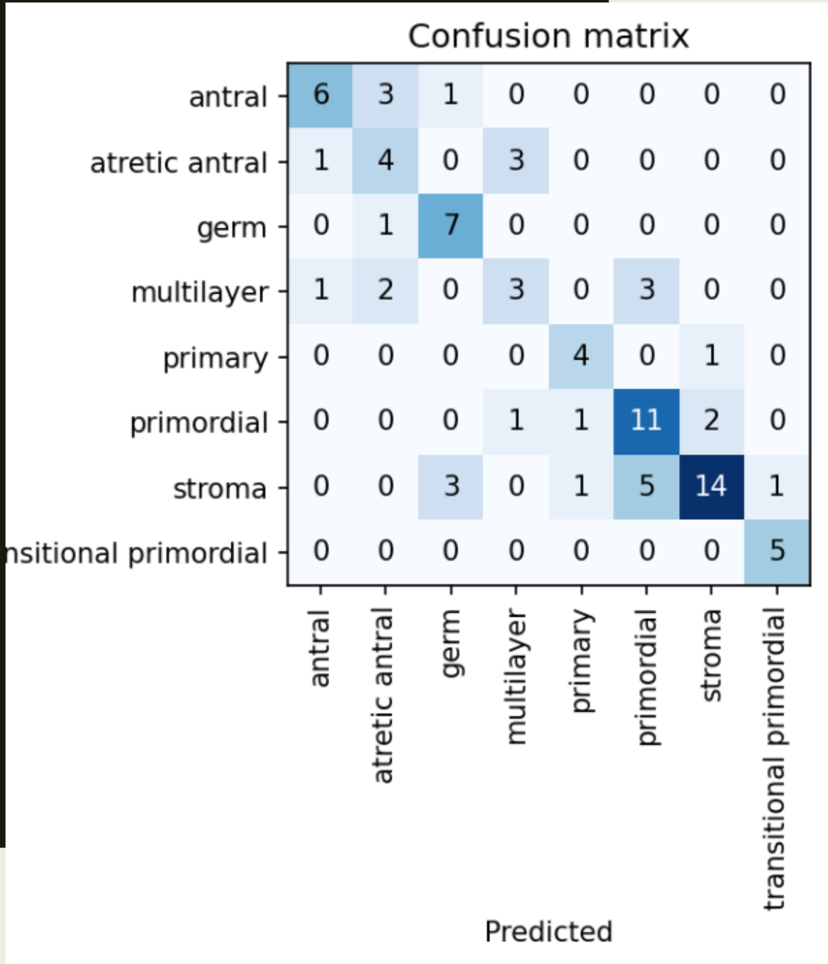
- Whole-slide images analyzed using QuPath
- Selected 3 slides for detailed manual annotation
- Automated tissue detection used to identify key regions
- Trained classifier to segment tissue types
- Standardized labeling improved consistency
- Generated small image “tiles” from annotated regions
- Tiles preserve detail and are ready for machine learning





OpenAI. (2026). **ChatGPT** (April 25 version) [Workflow generation of CNN with WSI Evaluation]. https://chat.openai.com/chat

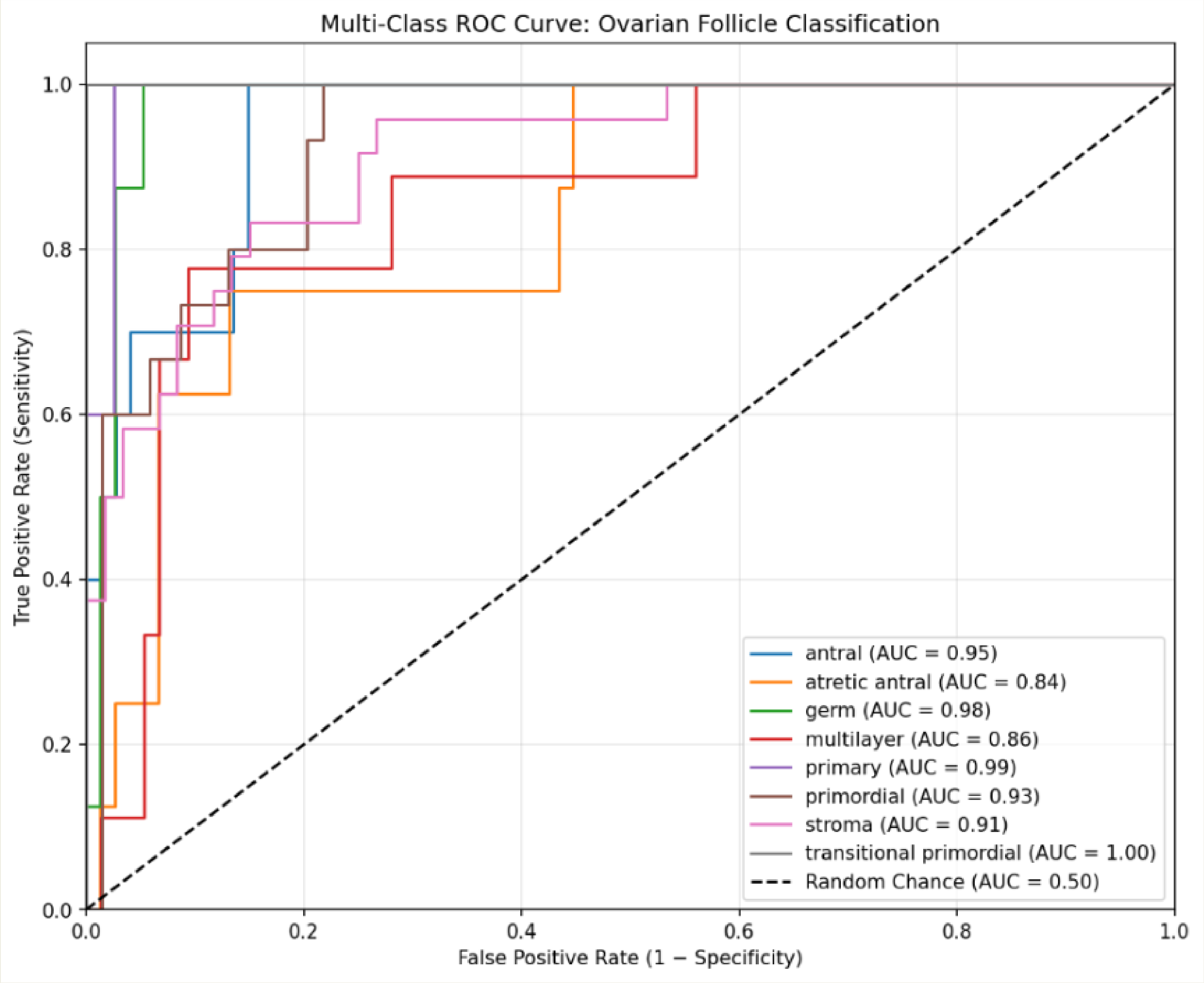
METHODOLOGY: CNN TRAINING AND EVALUATION

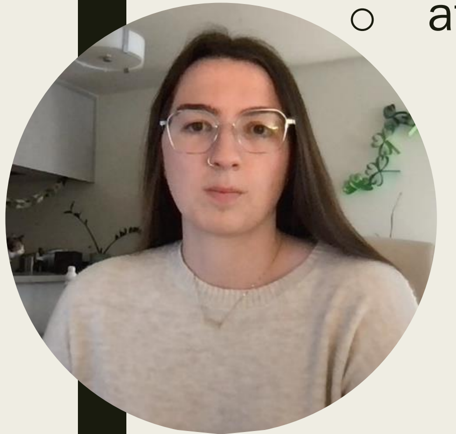


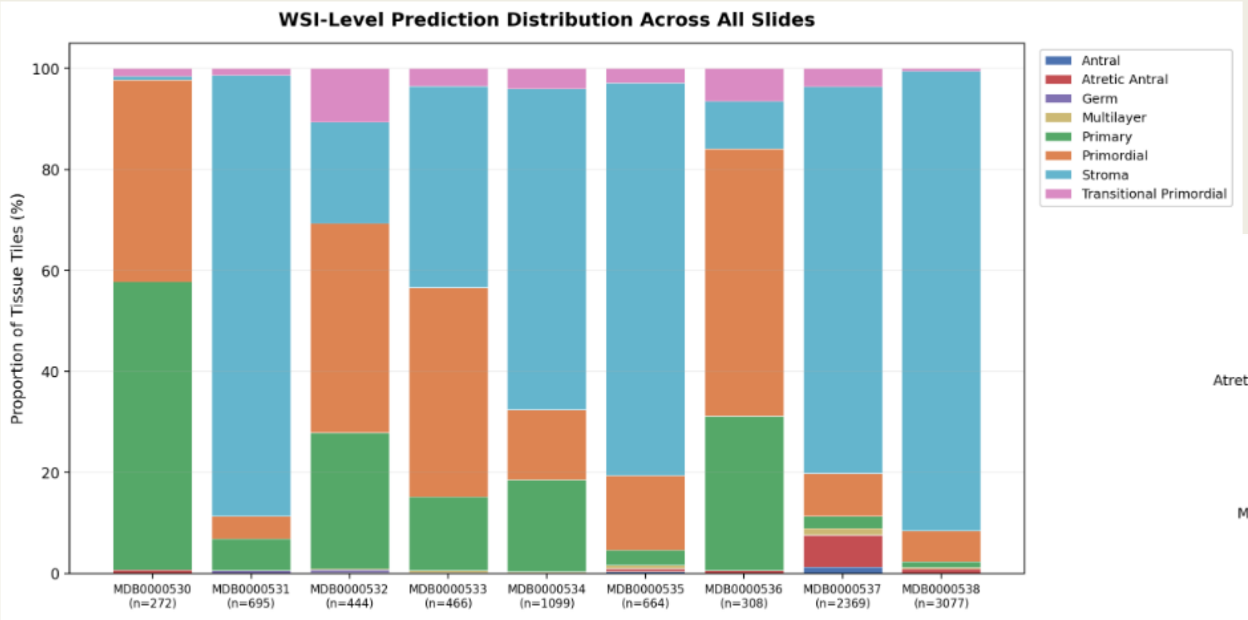
CNN CHECKPOINT VALIDATION AND CONFUSION MATRIX

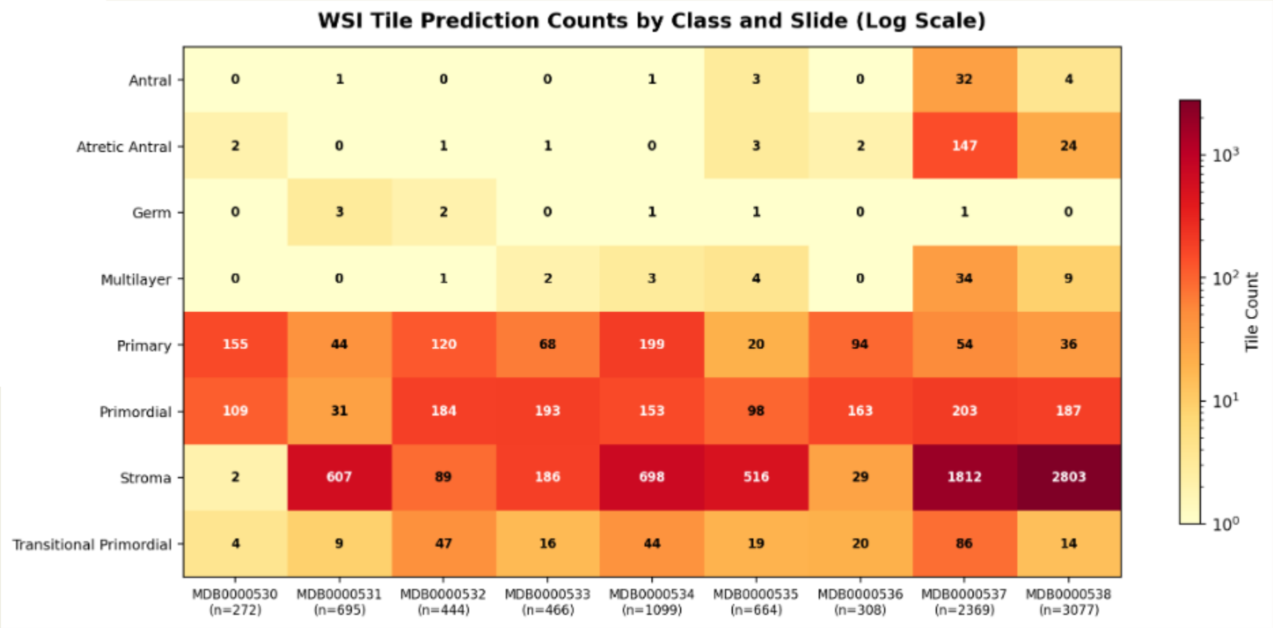
Receiver Operating Characteristic (ROC) Performance

- Mean Area Under Curve (AUC) = 0.93
- Best classes:
  - transitional primordial 1.00
  - primary 0.99
  - germ 0.98
- Lower-performing classes:
  - multilayer 0.86
  - atretic antral 0.84









WHOLE-SLIDE IMAGE ANALYSIS

Discussion

- **Model Performance and Interpretation**
  - Classification performance: moderate (ResNet34)
    - Best validation error: 26% on feature extraction phase
  - Fine tuning and overfitting
  - Reduced performance on test set at 32% error = Limited generalization
  - Strong performance on well-defined classes such as primordial and germ
  - Lower performance on morphologically complex classes such as stroma, multilayer and atretic antral

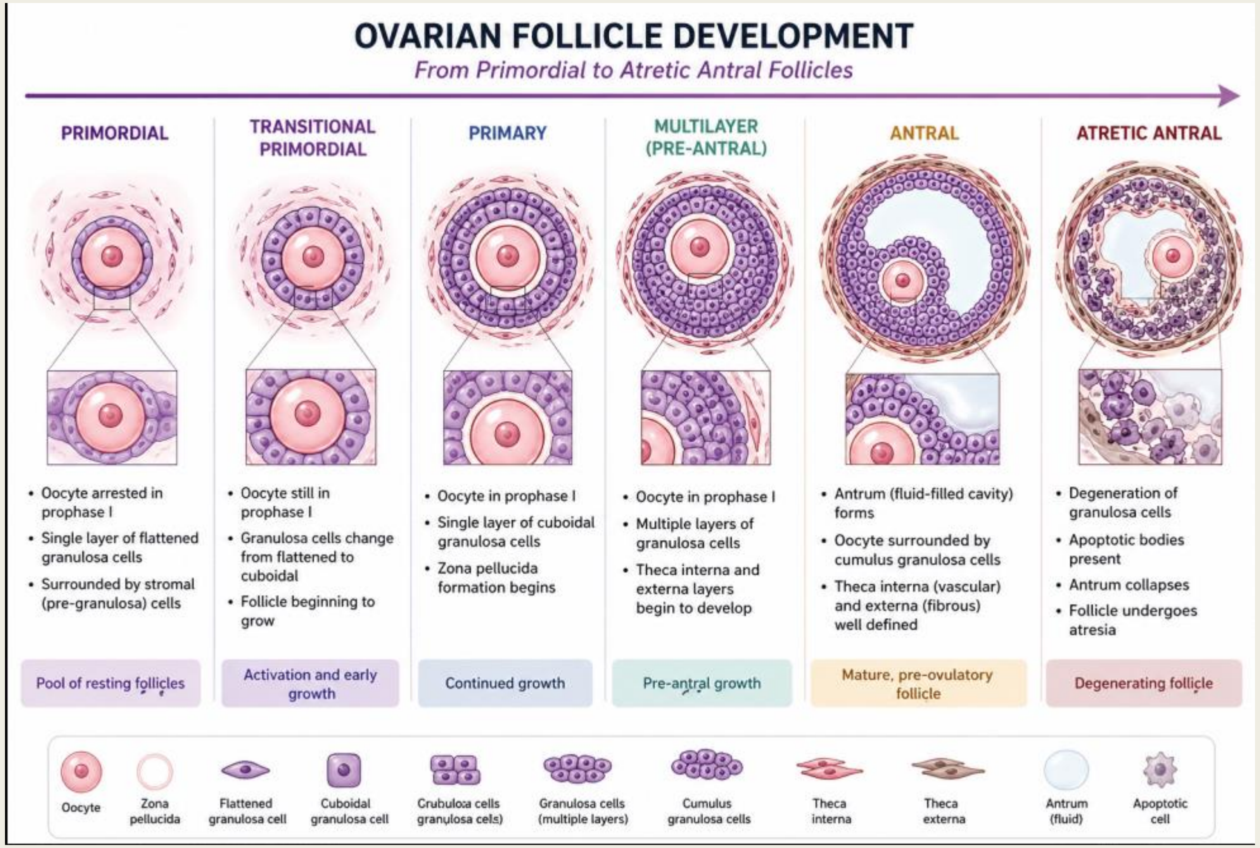
Discussion

Relevance

- Model captured biologically meaningful age-related patters
  - *Adult specimen slides = antral and atretic follicles*
  - *Younger specimen slides= dominated by early-stage follicles*
- Model demonstrated potential for automated histological analysis in H. glaber

Limitations & Implication

- Small dataset
- Class imbalance and annotation constraints
- Tile-based approach limits spatial precision
- Trade off (DL vs ML): efficiency vs anatomical detail
- Future improvements can consider combining segmentation(identify the exact boundaries of each follicle) and classification models with larger, more detailed annotated datasets to improve spatial precision, class accuracy, and overall generalization.



OpenAI. (2026) ChatGPT(April 26 version) [follicle_classes]. http://chat.openai.com/chat

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